

**ASSIGNMENT 1: Fourier Spectrum of Coherent Wavefield**

Due: Monday, October 13

**Objective:** The main objective of this homework assignment is to visualize the spectral distribution of 2-D coherent wavefield patterns over a planar circular aperture. This leads you to

- (i) the generation of the coherent wavefield,
- (ii) observations of the spectral distributions of coherent wavefield, and
- (iii) understanding of the concept of phase-only data acquisition.

**Part A: the wavelength**

- (1) Consider a point source at the origin,  $(x_o, y_o) = (0, 0)$ . This centered point source produces a 2-D coherent wavefield pattern in the form of

$$h(x,y) = (j\lambda_o r)^{-1/2} \exp(j2\pi r/\lambda_o)$$

where  $r = (x^2 + y^2)^{1/2}$  and  $\lambda_o$  is the operating wavelength. Generate the resultant 2-D wavefield pattern for a region within the radius of  $30\lambda_o$ . For simplicity, sample the 2-D wavefield uniformly, in both directions, with the sample spacing

$$\Delta x = \Delta y = \lambda_o / 4.$$

Evaluate the 2-D Fourier spectrum of the coherent wave-field with a  $512 \times 512$  FFT, and plot the amplitude of the 2-D spectrum. (Remember to zero out the data points around  $r = 0$ .)

- (2) Repeat Part (1) with 3 active point sources, located at  $(x_n, y_n)$ ,  $n = 1, 2, \dots 3$ .

|   | <i>scatters</i> | <i>scatter locations</i>        |
|---|-----------------|---------------------------------|
| 1 | $(x_1, y_1)$    | $(0, +15 \lambda_o)$            |
| 2 | $(x_2, y_2)$    | $(-12 \lambda_o, -9 \lambda_o)$ |
| 3 | $(x_3, y_3)$    | $(+12 \lambda_o, -9 \lambda_o)$ |

The resultant wavefield pattern over the aperture region is the superposition of 3 coherent waveform patterns.

- (3) Now consider a different case that each of these 3 active sources is generating an independent coherent wavefield pattern, with a different operating wavelength  $\lambda_n$ , where

|   | <i>scatters</i> | <i>scatter locations</i>        | <i>wavelengths</i>          |
|---|-----------------|---------------------------------|-----------------------------|
| 1 | $(x_1, y_1)$    | $(0, +15 \lambda_o)$            | $\lambda_1 = \lambda_o$     |
| 2 | $(x_2, y_2)$    | $(-12 \lambda_o, -9 \lambda_o)$ | $\lambda_2 = \lambda_o / 2$ |
| 3 | $(x_3, y_3)$    | $(+12 \lambda_o, -9 \lambda_o)$ | $\lambda_3 = 2 \lambda_o$   |

Again, the overall wavefield pattern over the aperture is the superposition of 3 waveform patterns. Sample the composite waveform in both directions, with the same sample spacing  $\Delta x = \Delta y = \lambda_o / 4$ . Evaluate the 2-D Fourier spectrum of the composite waveform with a  $512 \times 512$  FFT, and plot the amplitude of the spectrum.

### Part B: the magnitude

- (4) Repeat the exercise by setting the magnitude of your data samples to a constant, which is known as the *phase-only technique*.

Report format:

1. Cover page
2. Figures
3. Summary: (comments based on your observations)
4. Appendix: (computer code)